

Andre Byukusenge

SOFTWARE ENGINEER

CONTACT

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PHONE
+250 741 522 244

LOCATION
Nyabihu, Rwanda

PORTFOLIO
byukusengeandre.vercel.app

GITHUB
github.com/Byukusenge-Andrew

SKILLS

PROGRAMMING
C / C++, Python, Java, Ruby, JavaScript, TypeScript

BACKEND
NestJS, Django, Spring Boot, REST APIs, JWT authentication

FRONTEND
React, Next.js, Angular, Three.js

DATABASES
PostgreSQL, MySQL

CLOUD
AWS EC2

AI / ML
QLoRA fine-tuning, GGUF and llama.cpp, RAG with FAISS

SECURITY
Network security, code review, vulnerability analysis

OTHER
IoT fundamentals

LANGUAGES

English	Expert
Kinyarwanda	Expert
French	Intermediate

PROFILE

Software Engineer across embedded systems, backend engineering and applied AI/ML, seeking an internship. Best represented by **Rights Navigator**, a legal AI assistant fine-tuned and quantized for fully offline, CPU-only use across four African jurisdictions. Experienced in production NestJS/PostgreSQL services, systems-level programming in C/C++, and cybersecurity work spanning code review and vulnerability analysis, with independent research into novel RAG indexing techniques. Delivered work for international clients and internal teams, and mentored junior developers at Rwanda Coding Academy.

EDUCATION

High School Diploma, Software Engineering & Embedded Systems 2026

Rwanda Coding Academy

- Studied software engineering together with embedded systems.
- Completed specialized cybersecurity training and placed 2nd in a nationwide cybersecurity competition.
- Took part in coding competitions, mentored junior students and joined the STEM club.

PROFESSIONAL EXPERIENCE

Software Developer · *Echo Sols* Aug 2025 – 2026

- Developed backend services using NestJS and PostgreSQL for production applications, ensuring high performance.
- Optimized database queries and API performance to improve response times and user experience.
- Deployed and maintained services on AWS EC2, ensuring reliable operations and scalability.

Documentation & Code Review Intern · *Evu Ventures* Feb 2026 – Apr 2026

- Wrote and maintained technical documentation for internal software products.
- Reviewed code contributions to help maintain quality and consistency across projects.

Freelance Software Developer · *Upwork* Jan 2025 – Nov 2025

- Delivered remote freelance projects for international clients, achieving high satisfaction ratings and repeat business.
- Built automation tools and backend scripts using Python and JavaScript, improving client workflows.
- Translated client requirements into reliable, production-ready solutions, ensuring on-time delivery and quality.
- Built JobMatch, an Angular/TypeScript frontend for a job-matching platform, for a freelance client.

PROJECTS

Rights Navigator — Python, Qwen3-4B, llama.cpp, FAISS, RAG

A legal assistant that works without internet, so it can run on ordinary CPU-only hardware (ADTC 2026 submission).

- Covers Rwanda, Kenya, Nigeria and Tanzania, with multilingual support (Swahili, Yoruba, French, Kinyarwanda).
- Fine-tuned a small model with QLoRA (training only a small set of added weights) and quantized it to GGUF so llama.cpp can run it locally.
- Built a two-stage RAG pipeline (retrieve relevant law text, then answer from it) with similarity-gated retrieval, so answers stay grounded in sources and the system says so when it has none.
- github.com/Byukusenge-Andrew/law

SMMS — React, Next.js, Python, Django

One dashboard for managing several social media accounts in one place.

- Covers account management, analytics, influencer collaborations, notifications and integrations.
- Django backend with a React / Next.js frontend.

Kredita — Mobile app, desktop app, hosted backend

A system for shops to manage customer debts and credit.

- Made of three connected parts: a mobile app, a desktop app and a hosted backend.
- Lets a shop keep track of what its customers owe and the credit it has given them.

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ACHIEVEMENTS

- 2nd place, nationwide cybersecurity competition
- FIRST Lego League Morocco — award for team collaboration, representing Rwanda
- Mentored junior developers in coding practices

CERTIFICATES

Cyberium Cybersecurity Training
2025 · Ethical Hacking & Defensive Strategies, network security labs

Ideation Workshop
Sep 2024 · Design Thinking & Creative Problem Solving

LINKS

Portfolio
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PROJECTS (CONTINUED)

Cachedb — C++
A Redis-like in-memory database written in C++.
- Multi-project / multi-database architecture.
- LRU, LFU and ARC caching policies decide which entries to evict when memory is full.
- HyperLogLog estimates the number of unique items using very little memory; Pub/Sub lets clients publish and subscribe to messages.

BUSGO — Java 17, Spring Boot 3, MySQL, JWT
Backend for booking bus tickets online.
- REST API covering companies, routes, stops, schedules, bookings and payments.
- JWT (token-based) authentication with role-based access control, in a layered controller / service / repository design.

Recovery Credit — Next.js
A web system for managing a credit portfolio and recovering overdue debts.
- Helps a lender keep track of its credit portfolio and of the debts that still need to be recovered.
- Built as a web application with Next.js (React).
- github.com/Byukusenge-Andrew/recovery-credit-6

ModelVerse — React, Three.js, TypeScript — interactive 3D visualization platform for rendering and manipulating complex models directly in the browser.

offline-edu — TypeScript — personal project exploring how to deliver education content in low-connectivity, offline environments.

EXTRACURRICULAR ACTIVITIES

Reading, Chess & CodeTech Clubs · *Rwanda Coding Academy, 2026*

Football Team Member · *Rwanda Coding Academy, Sep 2024 – 2026* — weekly training sessions and matches.

RCA Brainiacs Mentor · *Rwanda Coding Academy, Sep 2024 – 2025* — mentored junior students in programming and software development.

STEM Club Member · *Rwanda Coding Academy, Sep 2024 – 2025* — organized and joined weekly STEM sessions on science and robotics.

Innovative Designer · *BlockPython Team, Nyanza, Jan – Aug 2023* — led robot design ideation for a robotics club competing in FIRST Lego League.

REFERENCES

Available upon request.